

Bachelor of Science in Computer Science & Technical Communication

**4-Year Hybrid Master Curriculum (Total: 8 Semesters with
Weekly Breakdown)**

Technology & Knowledge Hub

Year	Semester	Course Code & Title	Weekly / Topic Breakdown	Status	Notes / Resources
1	1	CS 1101 + MIT 6.0001	Week 1: Intro to Computing & Python Setup	 Not Started	MIT Syllabus / Python.org
1	1	CS 1101 + MIT 6.0001	Week 2: Variables, Data Types & Operators	 Not Started	
1	1	CS 1101 + MIT 6.0001	Week 3: Conditional Statements & Logical Ops	 Not Started	
1	1	CS 1101 + MIT 6.0001	Week 4: Loops & Iteration	 Not Started	
1	1	CS 1101 + MIT 6.0001	Week 5: Functions, Parameters & Scope	 Not Started	
1	1	CS 1101 + MIT 6.0001	Week 6: Strings Manipulation & File I/O	 Not Started	
1	1	CS 1101 + MIT 6.0001	Week 7: Compound Data (Lists, Tuples, Dicts)	 Not Started	
1	1	CS 1101 + MIT 6.0001	Week 8: Debugging Strategies & Intro to OOP	 Not Started	
1	1	CS 102	Week 1-4: HTML5 Structural Tags & CSS3 Box Model	 Not Started	GitHub Hosting
1	1	CS 102	Week 5-8: Responsive Design & Git/GitHub	 Not Started	
1	1	ENG 101	Week 1-4: Reading Technical	 Not Started	Tech English

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			Documentation		
1	1	ENG 101	Week 5-8: Code Documentation & Academic Grammar	 Not Started	
1	2	CS 1102 + OSSU	Week 1-2: JVM Architecture & Control Structures	 Not Started	Java Fundamentals
1	2	CS 1102 + OSSU	Week 3-4: Methods, Arrays & OOP Principles	 Not Started	
1	2	CS 1102 + OSSU	Week 5-6: Inheritance, Polymorphism & Interfaces	 Not Started	
1	2	CS 1102 + OSSU	Week 7-8: Exception Handling & Java I/O	 Not Started	
1	2	MATH 1302 + MIT 6.042J	Week 1-2: Propositional Logic & Truth Tables	 Not Started	Discrete Math
1	2	MATH 1302 + MIT 6.042J	Week 3-4: Set Theory & Functions	 Not Started	
1	2	MATH 1302 + MIT 6.042J	Week 5-6: Introduction to Proofs & Counting	 Not Started	
1	2	MATH 1302 + MIT 6.042J	Week 7-8: Graph Theory & Boolean Algebra	 Not Started	
1	2	ENG 102	Week 1-4: Peer-Assessment Review Skills	 Not Started	

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1	2	ENG 102	Week 5-8: Structuring Technical Posts	 Not Started	Academic Integrity
2	3	CS 2204 + Nand to Tetris	Week 1-2: Binary Code, Logic Gates & HDL	 Not Started	Systems
2	3	CS 2204 + Nand to Tetris	Week 3-4: ALU Design, Registers & RAM	 Not Started	
2	3	CS 2204 + Nand to Tetris	Week 5-6: CPU, Memory & Instruction Set Architecture	 Not Started	
2	3	CS 2204 + Nand to Tetris	Week 7-8: Assembler Design & VM Architecture	 Not Started	
2	3	CS 2203 + OSSU	Week 1-2: DBMS Introduction & Relational Model	 Not Started	SQL Mastery
2	3	CS 2203 + OSSU	Week 3-4: Basic & Advanced SQL Queries	 Not Started	
2	3	CS 2203 + OSSU	Week 5-6: Database Normalization & Transaction	 Not Started	
2	3	CS 2203 + OSSU	Week 7-8: NoSQL Overview & DB Connectivity	 Not Started	
2	3	ENG 201	Week 1-8: README.md Files & Swagger API Docs	 Not Started	Tech Writing

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2	4	CS 1103 + OSSU	Week 1-2: Collections Framework & Multithreading	☐ Not Started	Adv Java
2	4	CS 1103 + OSSU	Week 3-4: GUI Basics & Network Programming	☐ Not Started	
2	4	CS 1103 + OSSU	Week 5-6: Linked Lists, Stacks & Queues	☐ Not Started	Data Structures
2	4	CS 1103 + OSSU	Week 7-8: Binary Trees & Event-Driven Prog	☐ Not Started	
2	4	CS 204 (Backend first)	Week 1-4: Node.js Architecture, Async & Express MVC	☐ Not Started	Server-side
2	4	CS 204 (Backend first)	Week 5-8: RESTful API Design, JWT & DB Integration	☐ Not Started	
3	5	CS 3303 + Princeton	Week 1-2: Big O Notation & Union-Find	☐ Not Started	Algorithms I
3	5	CS 3303 + Princeton	Week 3-4: Sorting Algorithms & Priority Queues	☐ Not Started	
3	5	CS 3303 + Princeton	Week 5-6: Binary Search Trees & Balanced Trees	☐ Not Started	
3	5	CS 3303 + Princeton	Week 7-8: Hash Tables & Optimization	☐ Not Started	

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3	5	CS 304 (React UI)	Week 1-4: ES6+ Features & Asynchronous JS	 Not Started	Modern Frontend
3	5	CS 304 (React UI)	Week 1-4: React.js Architecture, Hooks & SPA Routing	 Not Started	
3	5	ENG 301	Week 1-8: Tech Slides & Delivering Demos	 Not Started	Public Speaking
3	6	CS 303 + Princeton	Week 1-4: Graph Search (DFS/BFS) & Shortest Paths	 Not Started	Algorithms II
3	6	CS 303 + Princeton	Week 5-8: String Algorithms & LeetCode Patterns	 Not Started	Interview Prep
3	6	CS 2301 + OSTEP	Week 1-2: Process Concept, CPU Scheduling & Memory	 Not Started	Operating Systems
3	6	CS 2301 + OSTEP	Week 3-4: Concurrency, Locks & Distributed Overview	 Not Started	
3	6	DevOps Core	Week 5-6: Docker Containerization & GitHub Actions	 Not Started	CI/CD Pipelines
3	6	DevOps Core	Week 7-8: Cloud Basics (AWS) & Deployment	 Not Started	Cloud Engineering
4	7	CS 4404 + Stanford	Week 1-4: OSI Model, IPv4/IPv6 Routing & TCP/UDP	 Not Started	Computer Networks

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4	7	CS 4404 + Stanford	Week 5-8: Distributed Systems, REST & Net Security	 Not Started	
4	7	CS 402	Week 1-4: Cryptography & Web Vulnerabilities	 Not Started	Cyber Security
4	7	CS 402	Week 5-8: IAM, Ethical Hacking & Cloud Security	 Not Started	
4	7	RES 401	Week 1-8: Literature Review & Proposals	 Not Started	Research Methodology
4	8	INT 401	Week 1-8: 3-6 Months Remote Internship / OpenSource	 Not Started	Global Experience
4	8	CAP 401	Week 1-8: Architecture Design, Cloud Deploy & Viva	 Not Started	Capstone Final Thesis
4	8	ENG 403 (Enhanced)	Week 1-4: Portfolio Building & Upwork Navigation	 Not Started	Global Freelancing
4	8	ENG 403 (Enhanced)	Week 5-8: Writing SOP/Motivation & Cold Emailing	 Not Started	Scholarship Ready